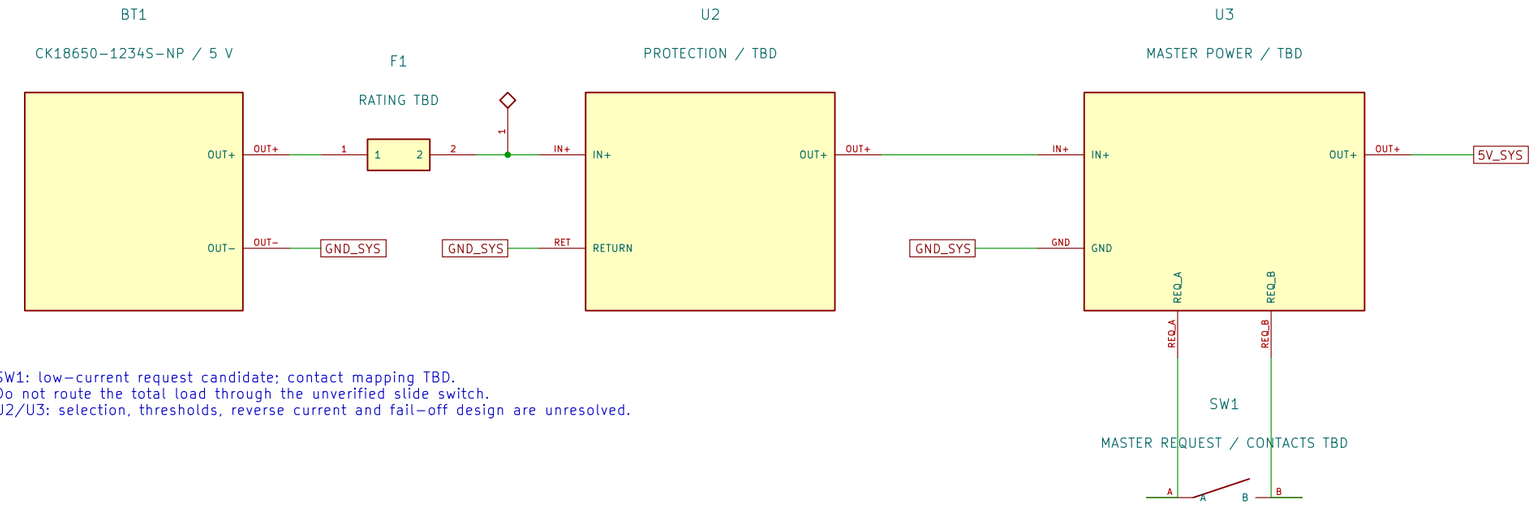
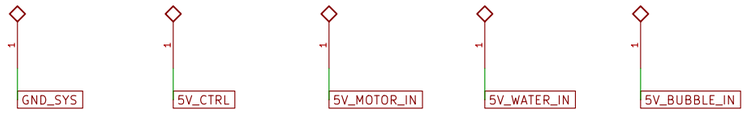
01 POWER INPUT AND DISTRIBUTION

A2 design candidate. Protection / switching blocks are NOT implemented circuits.



SW1: low-current request candidate; contact mapping TBD.
Do not route the total load through the unverified slide switch.
U2/U3: selection, thresholds, reverse current and fail-off design are unresolved.



BATTERY: output polarity / continuous current / protection not verified. Charging port excluded.

PWR_FLAG marks intended supply paths for ERC only; it is not measured power evidence.



OPEN JP1 for USB service. Do not power USB and 5Vin together.



SW2: ACTUATORS ONLY; controller remains powered (confirmed).
Device rating TBD; no MCU required to open. Re-arm / stored energy unverified.

All four action groups may run concurrently.
Fuse / wire / connector ratings require a measured input budget.

Ultrasonic 3V3 interface

文件: ultrasonic.kicad_sch

ESP32-S3 and BOOT

文件: control.kicad_sch

Motor drive

文件: drive.kicad_sch

Water and bubbles

文件: effects.kicad_sch

Reserved ignition LV

文件: ignition-lv.kicad_sch

Pin IDs identify signals, NOT physical header positions
Retain existing ESP32-S3 board; TBD items are unresolved
MODULE INTERCONNECTION SCHEMATIC - NOT FOR FABRICATION

CyberTrack

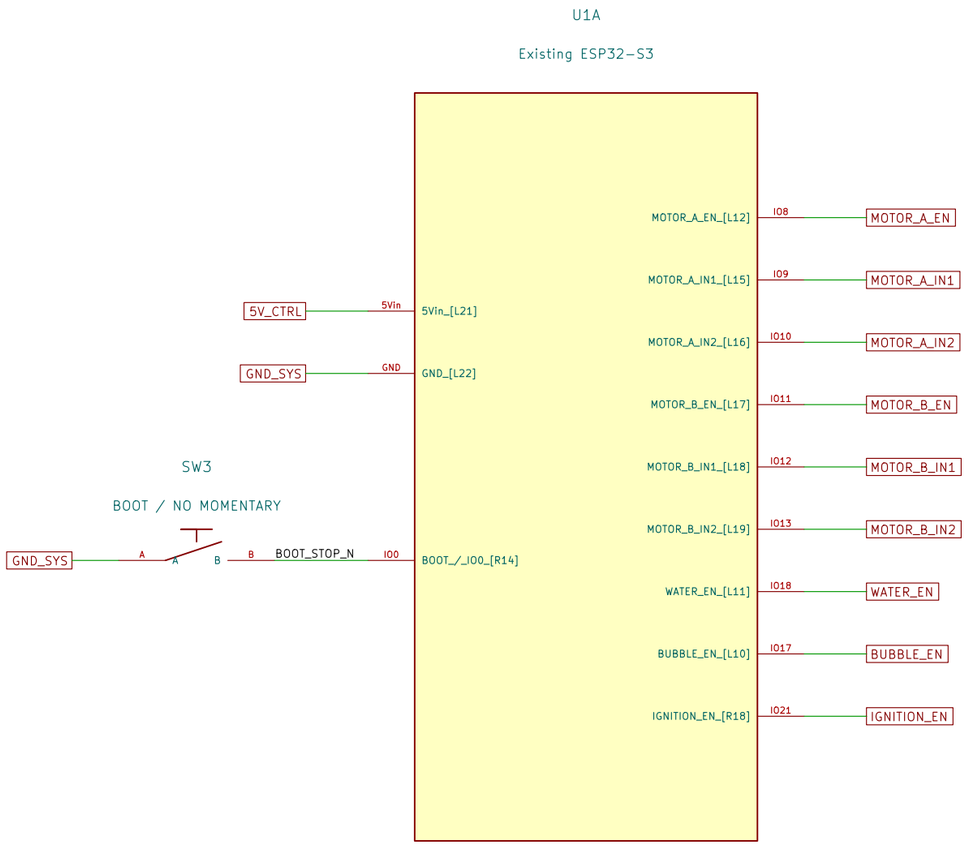
Sheet: /
File: CyberTrack.kicad_sch

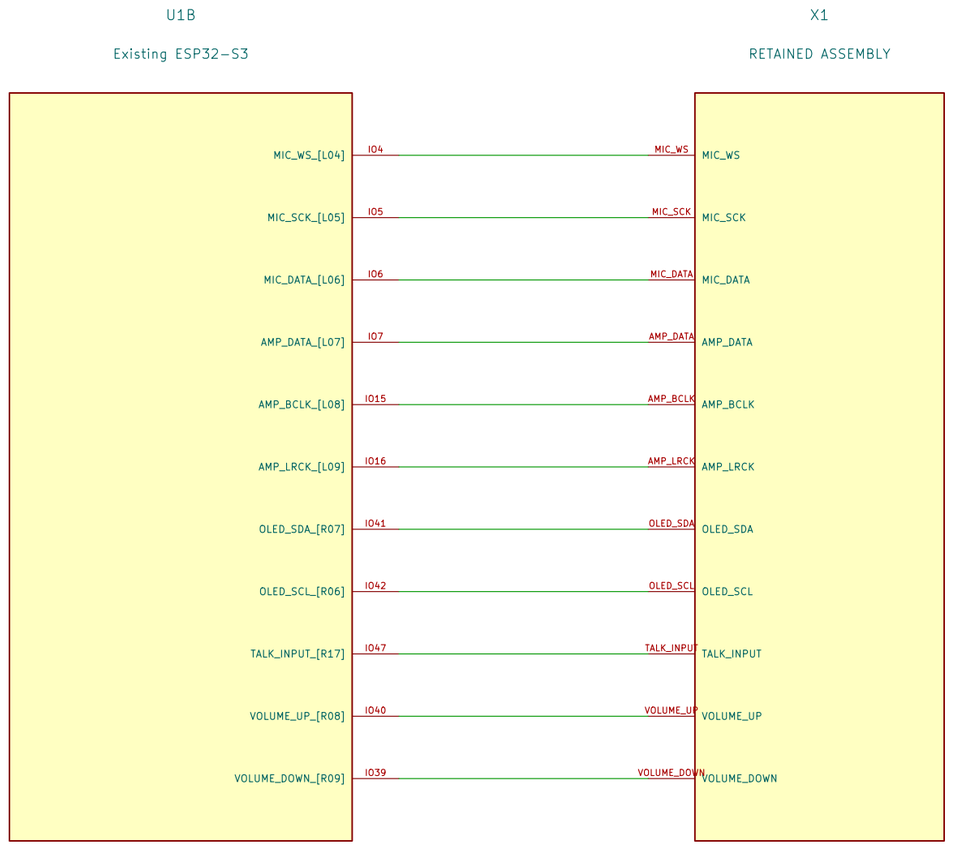
Title: CyberTrack / Power and isolation

Size: A3	Date: 2026-10-30	**Rev: A2**
KiCad E.D.A. 10.0.7		Id: 1/6

02 EXISTING DEVELOPMENT BOARD AND RETAINED PERIPHERALS

U1A / U1B / U1C are the SAME board. U1C is on sheet 6. USB, regulator and reset remain inside U1.





BOOT: dry contact to system ground; parallel to on-board BOOT.
It stops software actions. It does NOT isolate actuator power.
Holding BOOT during reset may select ROM download mode.
No external pull-up / RC values are frozen here.

X1 is a LOGICAL interface, not a new 11-pin connector.
Existing mic / amplifier / OLED / buttons retain their wiring.
Peripheral power and auxiliary pins are not reconstructed.
Do not infer module VCC or speaker grounding from this sheet.

PIN ID: IO8 = GPIO8; [L12] = photo-based header position.
Component side facing you, antenna UP, USB DOWN.
L/R rows counted 1-22 from antenna to USB; verify continuity.
GPIO1 / GPIO2 ultrasonic interface: sheet 6; no encoder / servo allocation.

Source: current action_outputs.h + cybertrack_tools.cc.
Retained peripherals: bread-compact-wifi/config.h.
Firmware pin identity does not prove physical wiring.
No PCB transfer; see parameter-confirmation.md.

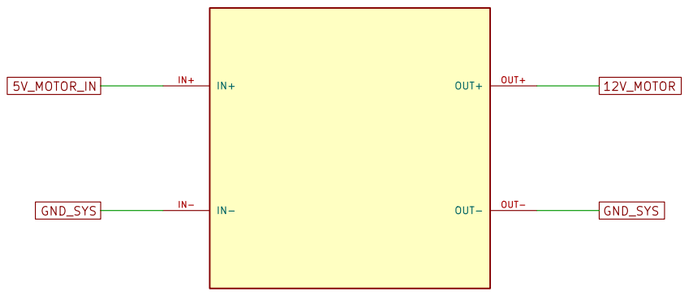
Pin IDs identify signals, NOT physical header positions
Retain existing ESP32-S3 board; TBD items are unresolved
MODULE INTERCONNECTION SCHEMATIC - NOT FOR FABRICATION
CyberTrack

Sheet: /ESP32-S3 and BOOT/		
File: control.kicad_sch		
Title: CyberTrack / ESP32-S3 and BOOT		
Size: A3	Date: 2026-10-30	**Rev: A2**
KiCad E.D.A. 10.0.7		Id: 2/6

03 EXTERNAL BOOST CONVERTER AND DUAL H-BRIDGE

U4

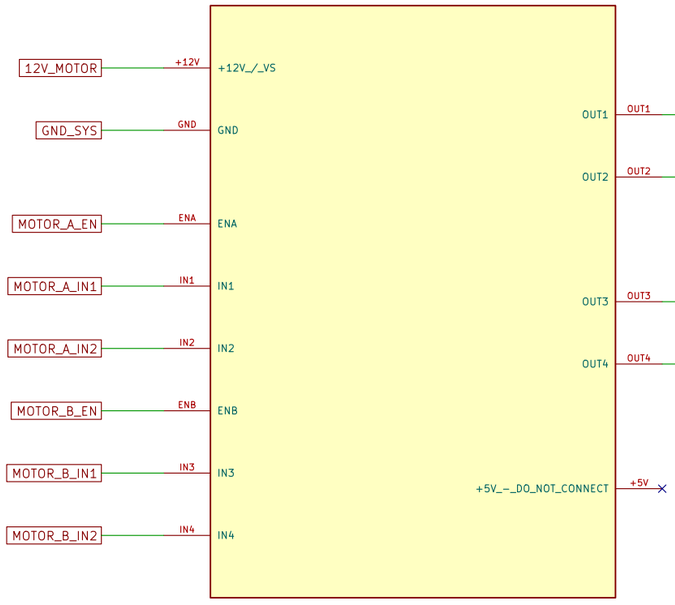
MT3608 / TARGET 12 V

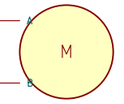


Remove ENA / ENB always-on jumpers.
Retain only the verified regulator-jumper configuration.
L298N +5V is intentionally unconnected.
12 V bus is not guaranteed motor-terminal voltage.

U5

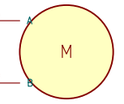
P13 / L298N





M1

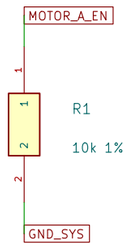
37R-520 / A

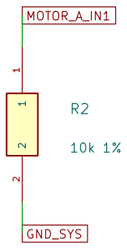


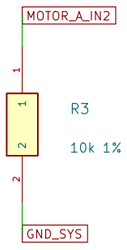
M2

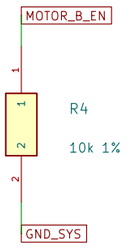
37R-520 / B

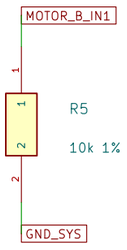
Current firmware: A = right; B = left.
Physical lead tracing / loaded tests still pending.
Encoder conductors are not connected.
Neither motor terminal is a fixed ground.

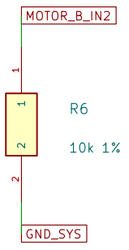












R1-R6: 10k, 1%, >=0.1 W; target 0603. At L298 inputs; actual module bias / reset state still need E05.
ST L298 Table 4 sizing: 10 uA x 10.1k = 0.101 V; added GPIO load at 3.3 V <= 0.433 mA (table conditions).
Photo control columns L to R: ENA, IN1, IN2, IN3, IN4, ENB; jumper contact identities need continuity check.
EN LOW = free-running stop, not active braking. Pull-downs cannot stop a GPIO stuck HIGH. Keep SW2 independent.
New-supply current, thermal and reverse-energy tests pending; see parameter-confirmation.md for limits.

Pin IDs identify signals, NOT physical header positions
Retain existing ESP32-S3 board; TBD items are unresolved
MODULE INTERCONNECTION SCHEMATIC - NOT FOR FABRICATION

CyberTrack

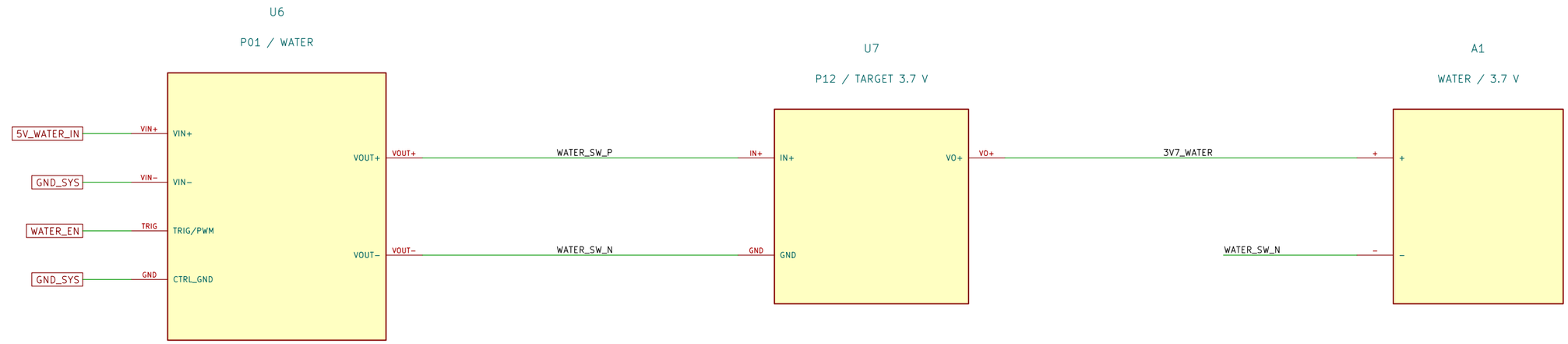
Sheet: /Motor drive/
File: drive.kicad_sch

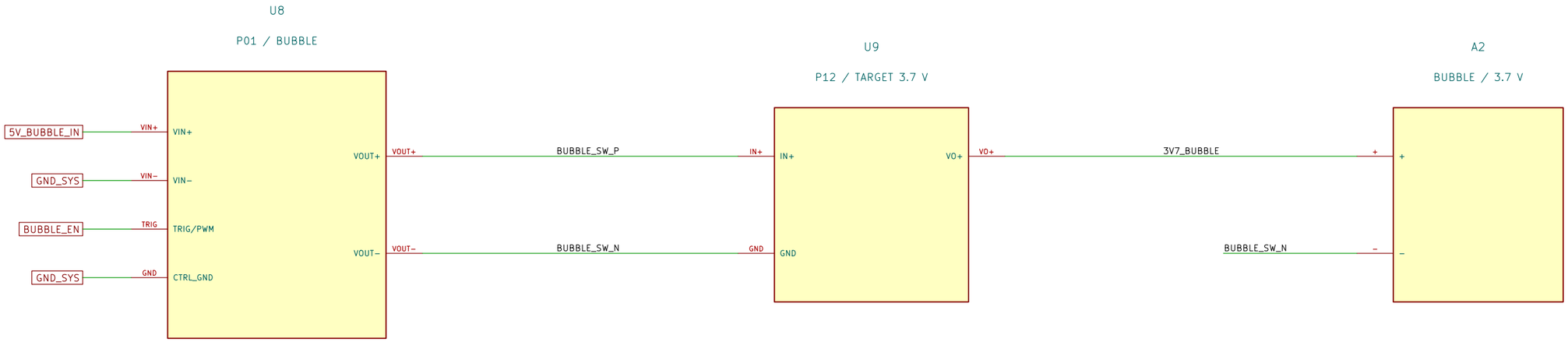
Title: CyberTrack / Motor drive

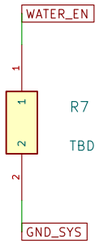
Size: A3	Date: 2026-10-30	**Rev: A2**
KiCad E.D.A. 10.0.7		Id: 3/6

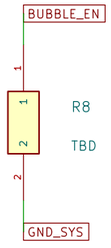
04 SEPARATE WATER AND BUBBLE POWER CHANNELS

MOS BEFORE buck converter. Keep each switched return separate from system ground and the other channel.









R7/R8 values TBD. P12 EN / trim retain existing settings, not modeled here.
P12 common GND is the local switched return, NOT GND_SYS.
JZ-MOS internal topology and 3.3 V trigger behavior require verification.
Do not bypass the switch with test-instrument ground or added wiring.
Buck capacitance may sustain output after MOS command goes LOW.

Pin IDs identify signals, NOT physical header positions
Retain existing ESP32-S3 board; TBD items are unresolved
MODULE INTERCONNECTION SCHEMATIC — NOT FOR FABRICATION
CyberTrack

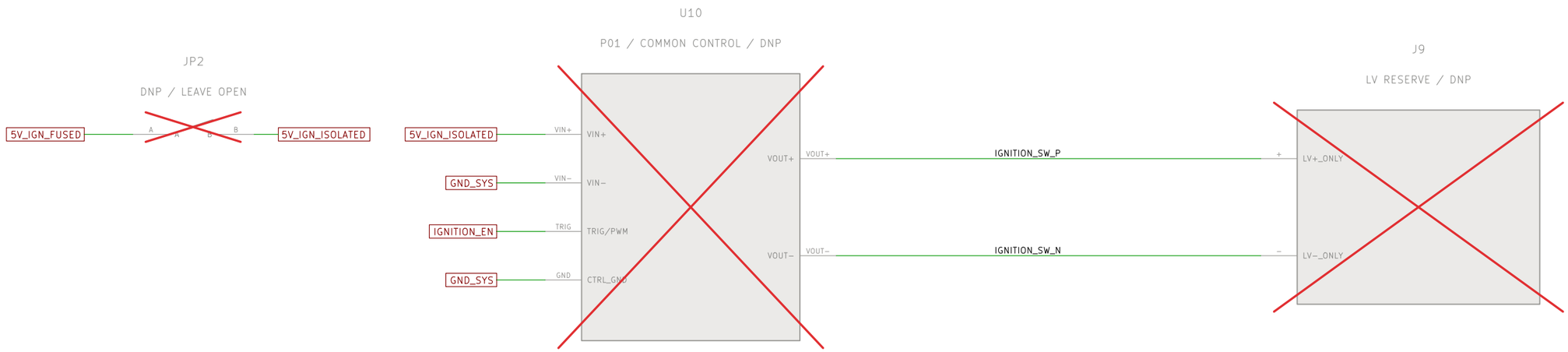
Sheet: /Water and bubbles/
File: effects.kicad_sch

Title: CyberTrack / Water and bubbles

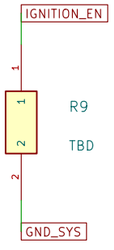
Size: A3	Date: 2026-10-30	**Rev: A2**
KiCad E.D.A. 10.0.7		Id: 4/6

05 IGNITION LOW-VOLTAGE INTERFACE - DISABLED

No ignition-module HV circuitry, electrodes or HV wiring is included. This sheet authorizes no energization.







Firmware: GPIO21, single action <= 5 s; cooldown >= elapsed software ON time.
These are software rules, NOT thermal protection or physical stop evidence.
R9 value, protection, input current, cut-off and EMC remain unresolved.
PWR_FLAG is an ERC declaration for the optional supply, not a populated link.

JP2 OPEN and output harness absent for initial validation.
U10/J9 marked DNP in this review assembly.
One control channel is reserved for the shared MOS requested for two modules.
Output pin numbers are logical +/- only; no connector or current rating is frozen.

Pin IDs identify signals, NOT physical header positions
Retain existing ESP32-S3 board; TBD items are unresolved
MODULE INTERCONNECTION SCHEMATIC - NOT FOR FABRICATION
CyberTrack

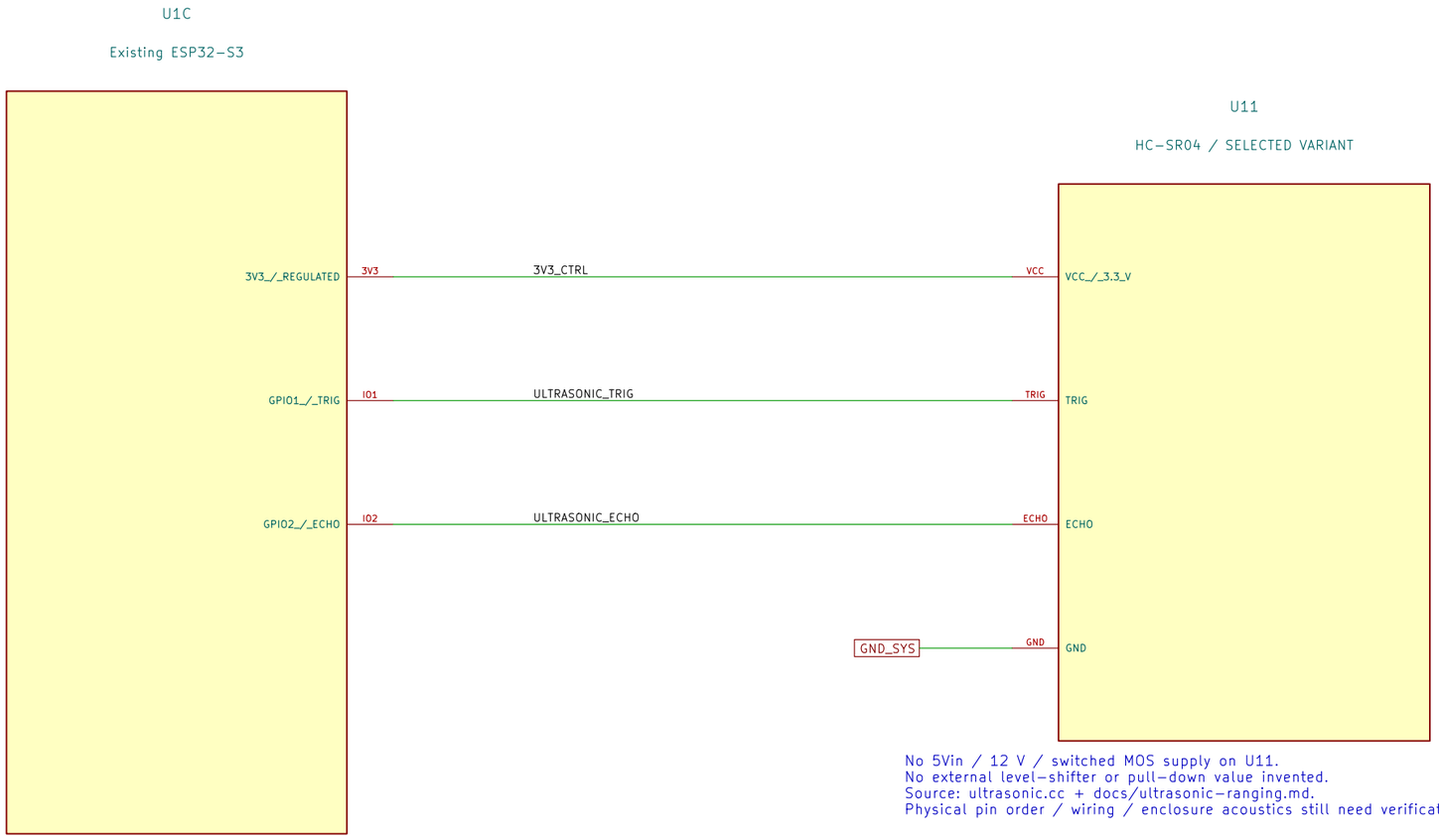
Sheet: /Reserved ignition LV/
File: ignition-lv.kicad_sch

Title: CyberTrack / Reserved ignition LV

Size: A3	Date: 2026-10-30	**Rev: A2**
KiCad E.D.A. 10.0.7		Id: 5/6

06 HEAD ULTRASONIC MODULE – CONTROLLER 3.3 V

U1C is part of the SAME existing ESP32-S3 board as U1A/U1B. No additional controller or regulator.



No 5Vin / 12 V / switched MOS supply on U11.
No external level-shifter or pull-down value invented.
Source: ultrasonic.cc + docs/ultrasonic-ranging.md.
Physical pin order / wiring / enclosure acoustics still need verification.

VCC: existing board 3V3 output. GND: system/controller ground.
GPIO1 -> TRIG: output, initially LOW, approximately 12 us pulse.
GPIO2 <- ECHO: input, firmware internal pull-down; direct connection scheme.
All four wires remain in the head; no added head-to-chassis signal wire.

USER-SELECTED 3.3 V SCHEME – NOT ELECTRICAL ACCEPTANCE
Verify module revision, 3V3 current margin, TRIG threshold and ECHO peak/off-state.
Do NOT change module VCC to 5 V while ECHO remains directly connected.
Signal IDs are NOT connector pad numbers; new header positions remain unassigned.

Pin IDs identify signals, NOT physical header positions
Retain existing ESP32-S3 board; TBD items are unresolved
MODULE INTERCONNECTION SCHEMATIC – NOT FOR FABRICATION
CyberTrack

Sheet: /Ultrasonic 3V3 interface/
File: ultrasonic.kicad_sch

Title: CyberTrack / Ultrasonic 3V3 interface

Size: A3	Date: 2026-10-30	**Rev: A2**
KiCad E.D.A. 10.0.7		Id: 6/6